Technical documentation for PakEmis 1.0.0rc69: greenhouse gas and air pollutant emissions, July 2015–June 2025

Rehan Ahmad[1]

[1]Pakistan Air Quality Initiative (PAQI), Karachi, Pakistan

Correspondence: Rehan Ahmad (rehan@pakairquality.com)

Abstract. This documentation describes the shared activity accounting, chemical conventions, source-specific calculations and reproduction interfaces underlying the national greenhouse-gas and air-pollutant dataset. The common core comprises ten fiscal years, July 2015–June 2025 (FY2015–16 to FY2024–25); FY2023–24 is the detailed reference year. Source-specific provisional estimates for FY2025–26 are kept separate. This document accompanies the national data-description article and the maintained sector notes and does not define another inventory total. Numerical summaries are read from the same released products as the main article. Appendix A lists every canonical data product.

1 Scope, period and record conventions

The native greenhouse gases are CO_2, CH_4 and N_2O. The core air panel is primary fine particulate matter ($PM_{2.5}$), black carbon (BC), nitrogen oxides (NO_x), sulfur dioxide (SO_2), carbon monoxide (CO), non-methane volatile organic compounds (NMVOC) and ammonia (NH_3). PM_{10}, total suspended particles (TSP), organic carbon (OC) and other species have narrower source coverage. Land use, land-use change and forestry are outside the inventory. Fluorinated gases, coal-mining fugitive emissions, several industrial processes and some treatment and combustion sources are listed as unestimated. A product that spans all modules is therefore not a complete category–species matrix.

A fiscal year runs from July to June and is identified by its start and end years; the FY2024–25 core endpoint is June 2025, not calendar 2025. A monthly publication through June 2026 can support a full FY2025–26 physical series when all required months and variables are present, whereas the Economic Survey 2025–26 reports several series for July–March only. The availability table records edition, covered months, revision status, variable, unit, held/public location and usability before a source is admitted to an extension.

The annual records identify the source, process, reporting category and calculation module separately. They retain native gas/species, mass convention, activity quantity and unit, activity period and provenance, factor unit and applicability, technology, control state, configuration, estimate status, temporal/spatial support and uncertainty model. Estimated zero, not applicable and not estimated are distinct statuses. Source coverage is supplied by process, species and year; configuration alternatives are never implicitly added to the reference.

NO_x is NO_2-equivalent mass, with NO-N converted by $46/14$. NH_3-N is converted to NH_3 by $17/14$, and N_2O-N to N_2O by $44/28$. Native pollutant masses never use a global warming potential. The separate AR5 and AR6 aggregates identify the selected fossil/biogenic methane convention (IPCC, 2013, 2021). BC and OC cannot be added to compatible $PM_{2.5}$ totals; nor can $PM_{2.5}$, PM_{10} and TSP be summed across nested definitions. Precursors are not combined into emitted ozone or total ambient $PM_{2.5}$.

2 Source ingestion and shared activity

2.1 Monthly minerals and source controls

Monthly mineral production from the Pakistan Bureau of Statistics (PBS) is read at its printed mineral, year and provincial grouping. The printed year block, not the workbook name, identifies the year; worksheets are selected by table schema and provenance; and a province row keeps the mineral group whose label and printed total enclose it. A

printed dash or an illegible cell is kept as a missing value rather than a zero, and a province absent from the source is not added.

Exported records are compared with independently transcribed source quantities: province–month entries, published national months, compatible annual totals and the expected producing provinces. A twelve-month profile is classed as national observed timing only when both its months and its producing-area coverage support that reading; otherwise the profile and its evidence class name the substitute. Mineral production supports the timing of the corresponding upstream oil and gas segments, while coal mine output does not establish when kilns or power stations burn coal. Because normalisation removes a uniform duplication, an unchanged profile cannot validate an exported activity table, so source controls, exported quantities and profiles are checked separately.

2.2 Coal mass, energy and consumer allocation

Station coal consumption is reported in kt (1 kt = 1 Gg) by the National Electric Power Regulatory Authority (NEPRA) (NEPRA, 2024). For station i and rank r,

$$Q_{i,t} = M_{i,t}\, NCV_{r(i),t}, \qquad E_{\mathrm{CO_2},t} = \sum_i Q_{i,t} EF_{\mathrm{CO_2},r(i)}, \tag{1}$$

where Q is TJ, M is kt and NCV is TJ Gg^{-1}. Thus the blend in kg CO_2 TJ^{-1} is weighted by heat, not by generation. The reference retains the documented rank-default NCVs. Reported coal used is distinct from purchases, receipts, stock movements and generation. Testing-period consumption is included where it is actually reported.

When a commissioning lane has observed generation but no printed mass, the reconstruction retains an interval derived from the documented station-intensity evidence. Its midpoint defines the conditional reference calculation; the endpoints carry no assigned probability or confidence level. A missing latest-year station transcription causes a failure rather than assigning its entire generation to a reconstructed lane. Older aggregate coal energy remains a comparator, and a year without a published aggregate energy value remains absent in that comparator.

The national coal account retains production, imports, exports and stock change where observed, and the consumer columns for public power, households, brick kilns and cement/other industry. Captive and K-Electric-system coal have no printed fuel quantity and remain, unsplit, inside the published cement and other industry column. If exports or stocks are not observed, the resulting discrepancy is not labelled a measured physical residual. The source records' coal origin, rank, moisture or calorific basis remain distinct.

The adopted industrial publication vintage is selected explicitly. For FY2024–25 the full-year table in the later Economic Survey assigns brick kilns and cement and other industry differently from the earlier provisional supplement, with the same combined activity. Both versions remain traceable. Substituting station consumption for a publisher's power allocation does not establish which non-power consumer used the difference. The unresolved quantity remains in the account and is not automatically emitted as kiln fuel.

Dated kiln counts are evaluated with the stated fuel rate and operating-season assumptions. The Punjab fleet cases retain their different observation dates and are scaled to the national total by the Punjab share of counted kilns; each count is compared with the booked allocation only in the fiscal year in which it was made, and the cases are not averaged. Negative supply-minus-use quantities under a case are retained as contradictions or unresolved source differences, not rounded away. Source-specific GHG and pollutant effects of these cases are reported separately from the conditional activity distribution.

2.3 Delivered gas, processes and transport fuel

The gas ledger distinguishes production, processing, utility delivery, chemical feedstock, process fuel and other combustion. Fertiliser feedstock is separated from combustion fuel where the source supports it. It is not included in the post-meter combustion-leakage base merely because a utility calls both quantities fertiliser deliveries. Methane releases are estimated with segment-specific activities and factors; national unaccounted-for gas is not used as a measured methane loss.

The shared gas deduction acts on the combustion activity to which the post-meter factor applies. A draw assigning additional methane release correspondingly changes the combusted fraction. This coupling prevents the same gas from being fully burned and lost as methane. It does not resolve all carbon differences between national gas statistics and source-specific process production.

Petroleum product sales retain customer-class and fuel distinctions. Retail diesel is partitioned between road use and off-road machinery; petrol attributed to residential generators is deducted from the quantity otherwise available to road transport. Operator fuel distinguishes railway traction from non-traction activity. Domestic aviation is separated from international bunkers and landing/take-off from cruise where the evidence permits. Electrical energy bought from another generator is not counted as a second combustion source. K-Electric's fuel-specific station record and purchased generation are therefore separate boundary questions. Its FY2024–25 gas and regasified liquefied natural gas (RLNG) coverage case converts gross-heat volumes to the station net-calorific basis before comparison or emissions calculation; oil and coal use their reported physical fuel quantities.

3 Agricultural nitrogen and household configurations

3.1 Census-consistent historical activity

Historical livestock levels are anchored to the adopted 2024 census (PBS, 2026a). For species a and proxy stock $P_{a,t}$,

$$H_{a,t} = H_{a,2024} \frac{P_{a,t}}{P_{a,2024}}. \tag{2}$$

The proxy supplies relative changes, not a second census level. It is the FAOSTAT livestock series, which reproduces the Economic Survey's projection from 1996–2006 inter-census growth rates, so the year-to-year change of every species, including the FY2024–25 growth, is a projected rate rather than an annual count; the 2024 census differs from that projection most for sheep. Beyond the last held proxy year, a species-specific geometric rate over the declared recent proxy window is applied to the census anchor, which is modelled extrapolation. The phase sensitivity uses the same bridge when assessing the six-month stock-to-fiscal alignment.

Crops use the full-year FY2024–25 tables of the Pakistan Economic Survey 2025–26 and historical fiscal editions; fertiliser product composition and offtake use the full-year Annual Fertilizer Review 2024–25 of the National Fertilizer Development Centre (NFDC) and historical editions (Ministry of Finance, 2024, 2026; NFDC, 2024, 2025). Cotton lint bales are converted to the seed-cotton mass required by the residue equations with the documented conversion. Crop production, area and fertiliser product composition retain distinct uncertainty and applicability. Fertiliser offtake is a supply quantity; it is not an observed application date or a direct measurement of field nitrogen loss.

3.2 Conserved management and field pathways

Nitrogen excretion for species a is

$$N_{\mathrm{ex},a} = H_a r_{N,a} M_a \, 365/1000, \tag{3}$$

where M_a is typical animal mass and $r_{N,a}$ is the daily nitrogen excretion rate per unit animal mass on the source's stated basis. The total is partitioned into grazing deposition, managed manure and diverted material. Destination assumptions do not change this partition into evidence that all diverted dung was combusted.

In the reference TAN formulation, incoming TAN is the species-specific fraction of the relevant nitrogen pool. At housing, yard or storage stage j,

$$\begin{aligned} L_{\mathrm{NH_3-N},j} &= TAN_j f_j, \\ TAN_{j+1} &= TAN_j - L_{\mathrm{NH_3-N},j} - L_{\mathrm{NO_x-N},j}. \end{aligned} \tag{4}$$

The adopted NH_3-N share of mixed volatilisation is 0.95, so the associated NO_x-N loss in management and grazing is $L_{\mathrm{NH_3-N}}(1-0.95)/0.95$ in the TAN formulation. This chemical attribution is not a measured Pakistani management or grazing factor. Other management losses remove TAN and organic nitrogen in proportion to their surviving pools where no evidence establishes another partition. Nitrogen reaching application is

$$\begin{aligned} N_{\mathrm{applied}} = N_{\mathrm{managed}} &- L_{\mathrm{NH_3-N}} - L_{\mathrm{NO_x-N}} \\ &- L_{\mathrm{leach}} - L_{\mathrm{N_2}} - L_{\mathrm{N_2O-N}}. \end{aligned} \tag{5}$$

Each fate is accounted once and closure is tested for every species/system and every uncertainty realisation.

Direct soil N_2O uses gross applied nitrogen on its applicable IPCC basis. Application volatilisation drives subsequent indirect N_2O and is not deducted again before direct N_2O. Grazing retains its distinct deposited nitrogen and factors. Synthetic NH_3 follows product-weighted nitrogen offtake and NH_3-N factors; the associated NO_x-N term is retained when constructing the indirect N_2O activity. Product factors already expressed as NH_3-N are not multiplied by an additional NH_3 share.

The alternative IPCC formulation applies its complete prefield and field volatilisation fractions to the corresponding pools. The reference upland, flooded-rice and grazing factor choices retain their source-defined applicability (IPCC, 2019). Synthetic nitrogen is partitioned to rice where supported; organic and residue inputs do not have a fully observed crop and irrigation allocation. Factor and allocation sensitivities are published, and agreement with FAOSTAT (FAO, 2025) is not used to select a factor.

A separate chemical-attribution diagnostic replaces the mixed-loss NH_3-N share by 1.0 and reruns manure fate, field inputs, NH_3, NO_x and native greenhouse gases for each of the four configurations and all core years. In the TAN case this removes inferred management and grazing NO_x-N, changes the surviving manure nitrogen and thereby changes later NH_3 and N_2O. Separately factored applied-nitrogen and burning NO_x are retained. In the IPCC formulation the same diagnostic repartitions fixed mixed volatilisation losses, including application, while retaining their total nitrogen loss. The diagnostic is neither a measured alternative nor a plausible bound and has no probability. Air-pollutant masses and native-gas/AR5/AR6 consequences occupy separately typed tables; other sources are held at their configuration's central values.

3.2.1 The nitrogen partition against the EMEP/EEA 2023 process factors.

`pakemis_nitrogen_no_pathways.csv` compares each agricultural NO_x pathway with the factor the EMEP/EEA 2023 Guidebook publishes for it (EEA, 2023). For each pathway it gives the nitrogen the factor applies to, the central NO_x and its implied NO_2 per kilogram of nitrogen, the guidebook factor and table, and the case NO_x.

Table 1. Structural configurations. The machine identifiers are join keys; the reference is the survey/TAN combination. No probabilities are assigned.

Identifier	Household dung	Nitrogen formulation
S_TAN	Survey fuel use	TAN chain
L_TAN	All diverted dung combusted	TAN chain
S_IPCC	Survey fuel use	Complete IPCC fractions
L_IPCC	All diverted dung combusted	Complete IPCC fractions

- Synthetic fertiliser, applied manure and grazing excreta take the 3.D Tier 1 factor for N applied in fertiliser, manure and excreta, $0.04\,\mathrm{kg}$ NO_2 per kilogram of N (95 % interval 0.005–0.104), in place of the central 0.007 kg NO-N per kilogram of applied N and the 5 % NO_x share of grazing volatilisation.

- Manure management takes the 3.B Tier 2 storage factor, $0.01\,\mathrm{kg}$ NO-N per kilogram of ammoniacal nitrogen entering solid storage or litter, in place of the 5 % share of every management stage's mixed loss; the guidebook gives no NO factor for housing or yards.

- Crop-residue and burning NO_x are retained, because the guidebook gives no residue factor.

The resulting case, `nitrogen_no_emep2023_process_factors` in `pakemis_air_structural_sensitivities.csv`, is carried through national totals for every year and configuration. It is a documented alternative factor set, not a bound and not a probability. It changes NO_x only, and the greenhouse-gas pools are unchanged. NO from applied nitrogen is computed on that nitrogen, as the guidebook does, outside the volatilisation pool.

3.3 Survey-based and diverted-dung household activity

The household reference estimates fuel use from quantities in the Household Integrated Economic Survey (HIES), household counts from the 2023 census and supported fuel sales (PBS, 2019, 2026b, 2023). Interpolation and extrapolation between surveys are labelled. The alternative burns all the material diverted from manure management in households; it is a destination assumption. Under that alternative the uncertain herd drives both diverted dung and household emissions in the same realisation. Volatile solids, dry solids including ash, air-dry fuel and moisture remain distinct mass bases.

Residential pollutant factors represent gross appliance emissions. No nationally justified outdoor escape fraction is assumed. Household emissions should therefore not be interpreted as outdoor atmospheric-model releases without an additional boundary treatment. Unresolved dung destinations remain in a residual material account, with no invented combustion or return-to-soil pathway.

4 Air-pollutant source equations and coverage

4.1 Stationary combustion and processes

Activity is shared with the greenhouse-gas account, but pollutant factors retain their applicable boiler, furnace, engine, turbine, kiln or process boundary. GHG cement process and fuel emissions remain separate; the cement air factor represents the integrated kiln stack and is applied once to clinker output. Other industrial process and combustion boundaries retain separate source classes. Gas combustion uses the quantity remaining after the shared post-meter leakage deduction. A controlled factor receives no second removal efficiency. A plant licence or EIA can establish installed or designed equipment without establishing actual operation. The reference admitted power oil/diesel SO_2 remains distinct from coal screening and other unresolved plant pollutants.

Sulfur-based calculations keep fuel sulfur, retention and removal as separate quantities. Brick-kiln SO_2 is calculated from fuel sulfur only, with an assumed retention of 0.25; sulfur from the clay is not estimated. The cement-kiln factor combines fuel and raw-meal sulfur with in-kiln retention and cannot be separated into those parts. Limited kiln-fuel samples constrain particular fuels and dates; they are not a nationally representative sampling distribution. Uncontrolled combustion endpoints are conditional calculations; they are not treated as bounds on actual emissions.

Cement, ammonia, steel, soda ash and other process rows retain only their admitted source boundaries. Unestimated urea finishing, nitric acid, lime, glass or other industrial sources remain in the coverage register. Flaring is separate from venting; the same feed or fuel flow is not added in energy, IPPU and fugitives as three independent inputs.

4.2 Road, rail and aviation

Road air emissions comprise the supported exhaust, evaporative, tyre/brake and road-surface wear calculations. Exhaust factors include the guidebook's average hot/cold treatment; the dataset does not supply an independently calibrated Pakistan cold-start model. Registration survival, vehicle age and class/fuel proxies constrain the transferred factors. CNG chemistry and vehicle-class substitutions are labelled where retained. Road resuspension is excluded, and road estimates do not silently include off-road machinery.

Fuel-sulfur policy dates define specification scenarios rather than pump-level compliance observations. Technology and factor endpoints are propagated into pollutant consequences independently of the fuel/CO_2 closure. Non-exhaust wear follows distance/vehicle activity rather than treating its emissions as a fixed combustion factor. Rail factors follow operator traction fuel and their technology assumptions; do-

mestic aviation retains its source-specific activity and phase boundaries.

Monthly resident-airline jet-fuel purchases are a proxy for domestic-flight activity: the purchaser and flight-geography boundaries differ. Aviation therefore carries proxy timing support rather than observed monthly domestic movements. The common temporal ensemble's customer-sales hindcast describes dispersion within that proxy profile; it does not quantify the unresolved transfer from customer uplift to domestic-flight timing. National petrol distribution similarly treats road-customer sales as a timing proxy for a broader activity account.

4.3 Agricultural and waste burning

Crop burning uses production, residue ratios, burned fractions, combustion completeness and species factors on their documented mass bases. Residue returned to soil, diverted fuel and field burning are accounted on compatible definitions. A source-specific factor does not imply that the burned fraction was annually observed.

Waste generation per person follows the eight-city national survey (Pak-EPA and EPMC, 1996) and is partitioned into collected disposal and uncollected material. The latter is partitioned into burning and unresolved fate. The first-order-decay state carries the disposal history forward, rather than treating an annual population update as an observation of annual waste-management change (IPCC, 2006; Kaza et al., 2018; WHO and UNICEF, 2023). Open-burning CH_4 and N_2O retain their source-specific wet/dry bases. The pollutant calculation uses the same burned dry mass as the native waste uncertainty replay. Waste origin geography does not establish the location of disposal, treatment or burning.

5 Monthly and geographic products

For an allocation with annual mass $E_{s,p,t}$ and monthly weights $w_{s,p,t,m}$, following the structure of sector-specific temporal profiles (Crippa et al., 2020),

$$E_{s,p,t,m} = E_{s,p,t} w_{s,p,t,m}, \qquad \sum_m w_{s,p,t,m} = 1. \tag{6}$$

The weights carry their own source and evidence classification. A twelve-month profile with incomplete producing-area coverage is not national observed activity. A flat constant-daily-rate profile weights actual month lengths and leap days, expressing unresolved seasonality. Fertiliser sales, field application, manure stages, evaporation and combustion can require different profiles despite their shared annual input. FY2024–25 crop-residue burning is timed and placed by its own VIIRS detections. The area-API record behind the earlier nine years ends in June 2024, so this year's burning windows come from NASA's keyless country-year files for 2024 and 2025, the same VIIRS S-NPP standard product for Pakistan only. Every country-file detection inside the earlier years' windows is in the area-API record. The annual burned mass is estimated independently of these detections, which only time and place it. The held crop district panel stops at FY2023–24. Crop-driven FY2024–25 district weights therefore reuse its reference shape and are separately marked as a forward proxy. Residential and waste district weights driven by population projections are marked as modelled demographic patterns, even when the household survey strata are from FY2024–25.

Annual administrative and grid products preserve the distinction between reporting and physical geography. Facility commissioning/closure periods are retained, and historical allocations state whether they use own-year evidence, a fixed proxy, a reference pattern or a substitute. This prevents a repeated location surface from being interpreted as observed district relocation. The primary package and alternative geographic products identify their respective boundaries, source licences and reporting masks. Cement works are placed at plant coordinates from the Global Cement and Concrete Tracker (CC BY 4.0) (Global Energy Monitor, 2026); coordinates derived from OpenStreetMap are used only in the separate ODbL companion.

Spatial closure is

$$E^{\text{national}}_{p,t,c} = \sum_g E^{\text{placed}}_{p,t,c,g} + E^{\text{unallocated}}_{p,t,c}, \tag{7}$$

for species p, year t and configuration c, on a compatible reporting boundary. Territories outside a statistical total are not silently included by a broader grid footprint. The maximum residual is reported at the product's actual numerical precision. Meaningful discrepancies are not absorbed into rounding.

The $0.05°$ grid stores native mass per cell and period. Flux density requires mass-unit conversion, the appropriate cell/country area and period duration. A country fraction already incorporated into allocated cell mass must not be applied again as though it were a flux. Atmospheric-model use additionally requires release height, speciation and source-boundary choices; these are not supplied merely by placing mass on a grid.

6 Uncertainty and evaluation

The annual conditional ensembles use common parameter identities, seeds and draw indices for greenhouse gases and air pollutants. Positive quantities use the specified mean-preserving distributions; physically bounded fractions use distributions matched to their stated conditional mean/variance where justified. The selected distributions are modelling assumptions. Their ensemble percentiles are not probabilities assigned to structural cases or empirical confidence statements about missing sources.

The nitrogen engine regenerates its material state and native $N_2O/NH_3/NO_x$ releases. Crop-residue burning sums

crop-specific burned masses after applying their corresponding production draws, then applies the common combustion-completeness and species-factor draws; the same crop-activity identities reach the air calculation. The waste engine reruns its bounded and positive parameters and tests fate closure before accepting a realisation. Fugitives retain their source/activity and factor-group bounded model; identity-seeded streams preserve paired parameters across years. Coal heat and CO_2 factors are replayed by rank. Other streams retain the existing parameter graph and its documented approximation. Unsupported management/equipment shares and control performance are not independently perturbed into physically impossible partitions.

Fertiliser NH_3 factors retain their product-specific central values and share one table-wide multiplier in the conditional ensemble. This dependence assumption is not a measured cross-product covariance; the published interval is conditional on it.

Two annual dependence cases retain persistent parameters or resample annual activity errors. Trend intervals use paired realisations. Waste-management parameters that are held across the activity history persist in both cases. The four structural configurations remain separate; kiln operating seasons, fuel sulfur specifications and unresolved controls are additional deterministic cases without probabilities. Allocation uncertainty and unestimated sources are not folded into parameter percentiles.

The conditional level and paired-trend tables include native CO_2, CH_4 and N_2O masses and separate AR5 and AR6 aggregates. An analytical longitudinal sensitivity table, which does not replay every native material flow, is published with an explicit model-basis field and does not replace the conditional intervals. Pollutant intervals retain their chemical-mass convention. Driver products are calculated separately for greenhouse gases and each pollutant; correlation diagnostics identify sensitivity and are not summed as variance attribution. Prefix and reproducibility tests check that a shorter run reproduces the same initial realisations and that output-row ordering does not create new physical uncertainty.

Evaluation first matches period, gas/species, mass convention, category and fuel boundary. A half-weighted blend of adjacent calendar years is labelled a fiscal approximation unless monthly observations permit exact fiscal aggregation. Comparison with national reporting uses the stated IPCC/CRT crosswalk, not an unqualified join on category text. Comparisons with global inventories identify shared activity statistics and factors.

For pollutants, fossil-only versus fossil-plus-biomass industrial comparisons separate a coverage difference before testing factors. Power coal screening is not ranked alongside central admitted emissions. Provincial and district tests use common polygons and retain unallocated mass. Timing checks use source-month controls or evidence not used to create the same profile. Geographic comparisons distinguish construction proxies from independent evaluation evidence. Satellite columns or aerosol observations require appropriate transport, chemistry and retrieval treatment; a simple concentration–emission correlation is not used as validation.

7 Data products, reproduction and access

The FY2023–24 covered aggregate is $444.6\,\mathrm{Mt}\ CO_2e$ on the AR5 basis. The same release gives $422.8\,\mathrm{Gg}$ primary $PM_{2.5}$, $816.6\,\mathrm{Gg}\ NO_x$ and $2670\,\mathrm{Gg}\ NH_3$. The main article tabulates all core pollutants and shows selected native-gas maps, trends, structural cases and matched-boundary comparisons. These summaries are read from released products through a generated fact file.

The product catalogue identifies annual source records, administrative summaries, monthly activity/emissions, grids, coverage/factor/control evidence, conditional uncertainty, structural cases and evaluation tables. Source release versions and overlapping revisions remain traceable. Query interfaces use the same periods, units and canonical configurations as downloads and retain aliases for legacy identifiers where applicable.

Reproduction from compiled activity tables is distinguished from re-extraction of primary sources. The build records state which level was run, the dependencies, input hashes and outcome. The deposit includes the numerical products, source code, metadata, plotted data and reproduction instructions. Primary source documents are held by the project and redistributed only where their terms permit. Geographic alternatives derived from OpenStreetMap data under the Open Database Licence (ODbL) form a separate companion package.

7.0.1 Evidence tables behind the source checks.

Each parsed monthly source ships its own denominators and controls, and rows that fail a printed total are kept unrepaired in a quarantine file rather than corrected: `pbs_minerals_parse_summary.csv` and `pbs_minerals_quarantine.csv` for PBS minerals, with `pakemis_minerals_panel_screening.csv` recording which mineral-years of the province panel are whole; `pbs_qim_parse_summary.csv` and `pbs_qim_quarantine.csv` for the Quantum Index of Manufacturing; `ocac_parse_summary.csv` and `ocac_direct_edition_quarantine.csv` for petroleum sales, with `ocac_edition_currency.csv` naming the edition each month is built on and `ocac_direct_edition_crosscheck.csv` comparing differenced months with OCAC's single-month editions; `nfdc_monthly_parse_summary.csv` and `nfdc_monthly_quarantine.csv` for fertiliser offtake; and `apcma_dispatch_parse_summary.csv` and `apcma_dispatch_quarantine.csv` for cement

despatch. Monthly shapes are compared across sources in `limestone_qim_cement_crosscheck.csv` and `apcma_qim_cement_crosscheck.csv`, the residential temperature profile against ERA5 in `era5_degree_day_crosscheck.csv` (timing, not level) and the rice calendar against FAO harvest windows in `fao_crop_calendar_crosscheck.csv`. For coal, `pakemis_coal_allocation_identity.csv` tests whether the consumer split is a measurement or a redistribution, `pakemis_power_coal_energy_adjudication.csv` asks whether NEPRA's aggregate coal energy and its per-station coal mass can describe the same fleet, `power_table3_hypotheses.csv` enumerates the candidate explanations without selecting one, and `power_toe_reconciliation.csv` lays out the NEPRA and K-Electric reconciliation by year and fuel. `pakemis_gas_flow_crosswalk.csv` classifies every delivered gas stream by the kind of flow it is. Power-plant controls are recorded in `pakemis_air_power_control_evidence.csv` as public design evidence, without inferring demonstrated operation. `pakemis_faostat_nitrogen_flows.csv` compares managed soils with FAOSTAT at the nitrogen flows, and `pakemis_nitrogen_partition_sensitivity.csv` gives the greenhouse-gas consequences of the all-NH_3 chemical-attribution diagnostic, with no probability or bound. `pakemis_wastewater_method_comparison.csv` sets domestic wastewater N_2O under the 2019 Refinement beside the 2006 single-equation method. `pakemis_primary_geography_licence_audit.csv` records each primary-geography dependency and the scope of the separate ODbL companion, and `pakemis_citation_deposit_audit.csv` lists every output path the manuscripts cite with whether the deposit contains it.

A dataset DOI and external review-access deposit are pending. Neither this documentation nor the local package claims journal submission, external publication or acceptance.

Appendix A: Canonical data products

The release ships 233 canonical products. Each is listed once below, by family, with the file name under `outputs/` and one line on what it holds. Descriptions are taken from the product catalogue, the file's own header or its declared role.

Semantic query layer

- `semantic/dim_activity_columns.csv`: Semantic dimension table: activity columns.
- `semantic/dim_category.csv`: Semantic dimension table: category.
- `semantic/dim_configuration.csv`: Semantic dimension table: configuration.
- `semantic/dim_evidence_class.csv`: Semantic dimension table: evidence class.
- `semantic/dim_gas.csv`: Semantic dimension table: gas.
- `semantic/dim_geography.csv`: Semantic dimension table: geography.
- `semantic/dim_gwp.csv`: Semantic dimension table: gwp.
- `semantic/dim_month.csv`: Semantic dimension table: month.
- `semantic/dim_notation_key.csv`: Semantic dimension table: notation key.
- `semantic/dim_period.csv`: Dimension tables: the closed vocabularies every fact row keys into.
- `semantic/dim_pollutant.csv`: Semantic dimension table: pollutant.
- `semantic/dim_stream.csv`: Semantic dimension table: stream.
- `semantic/fact_activity_aviation.csv`: Semantic fact table: activity aviation.
- `semantic/fact_activity_crops.csv`: Semantic fact table: activity crops.
- `semantic/fact_activity_fertiliser.csv`: Semantic fact table: activity fertiliser.
- `semantic/fact_activity_fuel_sales_area.csv`: Semantic fact table: activity fuel sales area.
- `semantic/fact_activity_fuel_sales_national.csv`: Semantic fact table: activity fuel sales national.
- `semantic/fact_activity_gas_network.csv`: Semantic fact table: activity gas network.
- `semantic/fact_activity_gas_supply.csv`: Semantic fact table: activity gas supply.
- `semantic/fact_activity_household_fuel.csv`: Semantic fact table: activity household fuel.
- `semantic/fact_activity_industry_fuel.csv`: Semantic fact table: activity industry fuel.
- `semantic/fact_activity_industry_production.csv`: Semantic fact table: activity industry production.
- `semantic/fact_activity_livestock.csv`: Semantic fact table: activity livestock.
- `semantic/fact_activity_power_monthly.csv`: Semantic fact table: activity power monthly.
- `semantic/fact_activity_power_plants.csv`: Semantic fact table: activity power plants.
- `semantic/fact_activity_railways.csv`: Semantic fact table: activity railways.

- `semantic/fact_activity_source_details.csv`: Semantic fact table: activity source details.
- `semantic/fact_activity_vehicles.csv`: Semantic fact table: activity vehicles.
- `semantic/fact_activity_waste_flows.csv`: Semantic fact table: activity waste flows.
- `semantic/fact_activity_waste_services.csv`: Semantic fact table: activity waste services.
- `semantic/fact_adm1_month.csv`: Semantic fact table: provincial (ADM1) month.
- `semantic/fact_adm1_year.csv`: Semantic fact table: provincial (ADM1) year.
- `semantic/fact_adm2_month.csv`: Semantic fact table: district (ADM2) month.
- `semantic/fact_adm2_year.csv`: Semantic fact table: district (ADM2) year.
- `semantic/fact_air_absence_screening.csv`: Semantic fact table: air-pollutant absence screening.
- `semantic/fact_air_adm1.csv`: Semantic fact table: air-pollutant provincial (ADM1).
- `semantic/fact_air_adm1_year.csv`: Semantic fact table: air-pollutant provincial (ADM1) year.
- `semantic/fact_air_adm2.csv`: Semantic fact table: air-pollutant district (ADM2).
- `semantic/fact_air_adm2_year.csv`: Semantic fact table: air-pollutant district (ADM2) year.
- `semantic/fact_air_annual.csv`: Semantic fact table: air-pollutant annual.
- `semantic/fact_air_conditional.csv`: Semantic fact table: air-pollutant conditional.
- `semantic/fact_air_lahore.csv`: Semantic fact table: air-pollutant lahore.
- `semantic/fact_air_panel.csv`: Semantic fact table: air-pollutant panel.
- `semantic/fact_air_source_classes.csv`: Semantic fact table: air-pollutant source classes.
- `semantic/fact_air_timeseries.csv`: Semantic fact table: air-pollutant timeseries.
- `semantic/fact_air_trend.csv`: Semantic fact table: air-pollutant trend.
- `semantic/fact_annual.csv`: Typed fact tables (annual, calendar, monthly, ADM1, ADM2, unallocated, uncertainty, completeness, transfers, evidence) whose values are the products'.
- `semantic/fact_annual_calendar.csv`: Semantic fact table: annual calendar.
- `semantic/fact_completeness.csv`: Semantic fact table: completeness.
- `semantic/fact_completeness_mass.csv`: Semantic fact table: completeness mass.
- `semantic/fact_evidence.csv`: Semantic fact table: evidence.
- `semantic/fact_ghg_conditional.csv`: Semantic fact table: greenhouse-gas conditional.
- `semantic/fact_ghg_trend.csv`: Semantic fact table: greenhouse-gas trend.
- `semantic/fact_monthly.csv`: Semantic fact table: monthly.
- `semantic/fact_spatial_basis_year.csv`: Semantic fact table: spatial basis year.
- `semantic/fact_transfer.csv`: Semantic fact table: transfer.
- `semantic/fact_unallocated.csv`: Semantic fact table: unallocated.
- `semantic/fact_uncertainty.csv`: Semantic fact table: uncertainty.
- `semantic/fact_uncertainty_series.csv`: Semantic fact table: uncertainty series.
- `semantic/semantic_release.json`: The semantic contract: release identity, the dimensions and their values, the defaults, the rounding and aggregation rules, the source digests.

Gridded products (NetCDF)

- `pakemis_air_grid_0p05_annual_l_ipcc_fy2015_16_fy2024_25.nc`: Annual and monthly covered air emissions.
- `pakemis_air_grid_0p05_annual_l_tan_fy2015_16_fy2024_25.nc`: Annual and monthly covered air emissions.
- `pakemis_air_grid_0p05_annual_s_ipcc_fy2015_16_fy2024_25.nc`: Annual and monthly covered air emissions.
- `pakemis_air_grid_0p05_annual_s_tan_fy2015_16_fy2024_25.nc`: Annual and monthly covered air emissions.
- `pakemis_air_grid_0p05_monthly_l_ipcc_fy2015_16_fy2024_25.nc`: Annual and monthly covered air emissions.
- `pakemis_air_grid_0p05_monthly_l_tan_fy2015_16_fy2024_25.nc`: Annual and monthly covered air emissions.
- `pakemis_air_grid_0p05_monthly_s_ipcc_fy2015_16_fy2024_25.nc`: Annual and monthly covered air emissions.
- `pakemis_air_grid_0p05_monthly_s_tan_fy2015_16_fy2024_25.nc`: Annual and monthly covered air emissions.

- pakemis_ghg_grid_0p05_annual_l_ipcc_fy2015_16_fy2024_25.nc: Annual native gas per cell for named structural configuration.
- pakemis_ghg_grid_0p05_annual_l_tan_fy2015_16_fy2024_25.nc: Annual native gas per cell for named structural configuration.
- pakemis_ghg_grid_0p05_annual_s_ipcc_fy2015_16_fy2024_25.nc: Annual native gas per cell for named structural configuration.
- pakemis_grid_0p05_annual_ref_fy2015_16_fy2024_25.nc: Ten fiscal periods at 0.05 degrees, three spatial views, three gases, in native mass.
- pakemis_grid_0p05_monthly.nc: The 0.05 degree monthly cube in native gas mass, with district and provincial views, for the reference year.
- pakemis_grid_0p05_monthly_ledger.nc: Configuration L on the 0.05-degree monthly grid the three gas layers, CO2 bit-identical to the reference cube.
- pakemis_grid_0p1_edgar_aligned.nc: The annual 0.05-degree cube aggregated 2x2 at the odd parity that tiles EDGAR's 0.1-degree lattice exactly.

Uncertainty and structural cases

- pakemis_air_absence_screening_by_year.csv: What the panel's typed absences plausibly hold, per period and species.
- pakemis_air_conditional_uncertainty.csv: Conditional pollutant intervals with shared nitrogen, waste and combustion inputs.
- pakemis_air_structural_sensitivities.csv: Source-specific structural cases with no scenario probabilities.
- pakemis_air_trend_uncertainty.csv: Paired-realisations changes under declared temporal dependence.
- pakemis_air_uncertainty_drivers.csv: Conditional uncertainty sensitivity by pollutant.
- pakemis_air_uncertainty_parameters.csv: Parameters and constraints used in pollutant ensembles.
- pakemis_air_uncertainty_split_by_year.csv: The two uncertainty halves reported separately per (period, configuration, pollutant).
- pakemis_bridge_configurations.csv: Records: bridge configurations.
- pakemis_ensemble_ledger_summary.csv: Configuration L on the same draw_id as the reference ensemble.
- pakemis_ensemble_spatial_conditionals.csv: Spatial alternatives as conditional ensemble blocks.
- pakemis_ensemble_summary.csv: A 10,000-draw prefix of the national Monte Carlo carried to months and provinces on one draw_id, with both closure identities holding inside every.
- pakemis_ghg_conditional_uncertainty.csv: Configuration-conditional native gases and separate GWP aggregates.
- pakemis_ghg_trend_uncertainty.csv: Paired native-gas and separate GWP conditional uncertainty.
- pakemis_ghg_uncertainty_drivers.csv: Paired native-gas and separate GWP conditional uncertainty.
- pakemis_ghg_uncertainty_parameters.csv: Paired native-gas and separate GWP conditional uncertainty.
- pakemis_joint_uncertainty.csv: Joint uncertainty by category and by the resolution of the figure being read.
- pakemis_longitudinal_prefix.csv: Legacy analytical sensitivity.
- pakemis_longitudinal_uncertainty.csv: Ten annual analytical bands under three temporal-dependence assumptions.
- pakemis_structural_trajectories.csv: Every registered structural axis in every panel year.
- pakemis_trend_uncertainty.csv: Draw-wise paired differences from the legacy analytical graph, restricted to sectors reported in both years.
- power_air_screening_fy2015_16_fy2024_25.csv: Power's uncontrolled screening scenario for the same ten years, shipped beside the covered panel and never inside it.

Evaluation and comparison

- pakemis_air_lahore_by_year.csv: Lahore district's air pollutants by sector and year.
- pakemis_btr_comparison.csv: PakEmis against Pakistan's own national submission.
- pakemis_builtup_comparison_adm2.csv: Records: builtup comparison district (ADM2).
- pakemis_comparison_crosswalk.csv: What each external comparison compares.
- pakemis_enteric_btr_bridge.csv: Enteric CH4 by species, this inventory (FY2020-21, REF_2023_24, AR5) against Pakistan's.
- pakemis_industry_edgar_divergence.csv: The engine's industrial series against EDGAR'S where A fiscal blend exists.
- pakemis_validation_catalog.csv: Per-catalog-row validation status.
- pakemis_validation_five_classes.csv: Every validation-catalogue entry filed under what the comparison can establish.
- pakemis_validation_five_classes_summary.csv: Counts of pakemis_validation_five_classes.csv by class.

- pakemis_validation_measured.csv: Measured validation, one row per check records each check read, how many it.
- pakemis_validation_scorecard.csv: The validation scorecard the (property x grain) space, one row.
- unsd_coal_supply_control.csv: UNSD energy balances - Pakistan primary coal and peat supply block, tj, per edition x calendar year.
- unsd_industry_combustion_control.csv: Independent 1.A.2 industrial-combustion control total from the UNSD.

Administrative and spatial records

- air_pollutants_adm1.csv: Legacy FY2023-24 provincial air layer of seven sectors.
- air_pollutants_adm2.csv: Legacy FY2023-24 district air layer of seven sectors (power rows are screening, labelled by estimate_kind).
- pakemis_adm1_by_year.csv: Ten fiscal years at ADM1 and ADM2, annually and monthly, under four configurations.
- pakemis_adm1_ledger_2024.csv: Configuration L at ADM1 grain, stream x gas x month x province.
- pakemis_adm2_by_year.csv: Records: district (ADM2) by year.
- pakemis_adm2_crosswalk.csv: Records: district (ADM2) crosswalk.
- pakemis_adm2_frame_2022.csv: The OCHA COD-AB 2022 ADM2 frame (160 units, CC BY-IGO) crosswalked to the canonical 152-unit set.
- pakemis_adm2_ledger_2024.csv: Configuration L at ADM2 grain, stream x gas x month x shape v2).
- pakemis_adm2_monthly.csv: District monthly greenhouse-gas records for the reference year, without intervals.
- pakemis_adm2_monthly_by_year.csv: Records: district (ADM2) monthly by year.
- pakemis_adm2_monthly_sector.csv: District monthly greenhouse-gas records by sector for the reference year, without intervals.
- pakemis_adm2_nesting_check.csv: Check that district agricultural allocations sum to their provincial controls.
- pakemis_adm2_sector_by_year.csv: Records: district (ADM2) sector by year.
- pakemis_adm2_stream_by_year.csv: The per-stream district vector for all ten periods.
- pakemis_adm2_stream_reference.csv: The reference configuration's per-stream ADM2 vectors, annual (Gg), all modules.
- pakemis_air_adm1_by_year.csv: Ten-year provincial air totals at anchor-stream grain, the cube's own province marginals.
- pakemis_air_adm2_by_year.csv: Ten-year district air totals on the canonical shape frame, aggregated per (configuration, sector, pollutant).
- pakemis_air_monthly_unallocated.csv: Records: air-pollutant monthly unallocated.
- pakemis_air_spatial_support_by_year.csv: Every air (period, configuration, anchor, pollutant) mass beside its weight-registry row.
- pakemis_air_unallocated_by_year.csv: The air mass no geometry could place below ADM1, typed per period and configuration (the aviation cruise class).
- pakemis_ghg_configuration_adm1.csv: Four configurations placed with own component masses.
- pakemis_ghg_configuration_adm1_monthly.csv: Four configurations placed with own component masses.
- pakemis_ghg_configuration_adm2.csv: All four configurations at ADM2.
- pakemis_ghg_configuration_adm2_monthly.csv: Four configurations placed with own component masses.
- pakemis_ghg_configuration_unallocated.csv: Unallocated native and GWP mass for each configuration and fiscal month.
- pakemis_grid_fallbacks.csv: Records: grid fallbacks.
- pakemis_grid_fallbacks_by_year.csv: Records: grid fallbacks by year.
- pakemis_grid_provincial_check.csv: Records: grid provincial check.
- pakemis_grid_stream_surfaces_by_year.csv: Records: grid stream surfaces by year.
- pakemis_kiln_allocation_adm2.csv: Records: kiln allocation district (ADM2).
- pakemis_power_grid_basis_change.csv: Comparison of the power geography under the alternative allocation bases.
- pakemis_road_length_adm2.csv: Records: road length district (ADM2).
- pakemis_road_links.csv: Records: road links.
- pakemis_road_technology_case.csv: Road exhaust PM2.5 and bc under technology-specific tier 2 factors the central.
- pakemis_road_weights_adm2.csv: Records: road weights district (ADM2).
- pakemis_spatial_basis_classes.csv: The spatial evidence vocabulary (spatial_time_basis), defined in plain words.

- pakemis_spatial_basis_shares.csv: Mass-weighted share of each period's emissions by spatial evidence class, national.
- pakemis_spatial_basis_shares_by_sector.csv: Mass-weighted share of each period's emissions by spatial evidence class, per IPCC reporting sector.
- pakemis_spatial_brackets.csv: Records: spatial brackets.
- pakemis_spatial_brackets_adm2.csv: Records: spatial brackets district (ADM2).
- pakemis_spatial_decomposition.csv: Records: spatial decomposition.
- pakemis_spatial_exceptions.csv: Allocations where the grid could not follow the declared spatial basis.
- pakemis_spatial_unallocated.csv: The mass with no defensible placement below ADM1, retained rather than spread.
- pakemis_spatial_unallocated_by_year.csv: Records: spatial unallocated by year.
- pakemis_spatial_weights_by_year.csv: Records: spatial weights by year.
- pakemis_stream_support_ledger.csv: One row per (period, stream).
- pakemis_unallocated_activity_controls.csv: Records: unallocated activity controls.

Monthly records

- pakemis_agriculture_crop_calendar.csv: Which FAOSTAT calendar year each fiscal crop cell actually is.
- pakemis_air_monthly.csv: Process-specific pollutant profiles with native mass and temporal evidence.
- pakemis_air_temporal_support.csv: Pollutant-specific temporal evidence shares.
- pakemis_fy2025_26_provisional_monthly.csv: Separate provisional source activity and emissions.
- pakemis_ghg_configuration_monthly.csv: Twelve process allocation months per source and fiscal year.
- pakemis_monthly_controls.csv: Records: monthly controls.
- pakemis_monthly_coverage.csv: What each month's timing rests on, one row per (month, sector, category, class).
- pakemis_monthly_coverage_summary.csv: Exactly 120 rows: what each month's timing rests on, and which sectors a month does not have at all.
- pakemis_monthly_data_requests.csv: Records: monthly data requests.
- pakemis_monthly_evidence.csv: Records: monthly evidence.
- pakemis_monthly_fy2015_16_fy2024_25.csv: 120 months, July 2015 to June 2025, every non-absent component resolved into twelve months closing exactly to its annual figure.
- pakemis_temporal_decomposition.csv: Records: temporal decomposition.
- pakemis_temporal_profiles.csv: Records: temporal profiles.

Coverage and evidence registers

- completeness.csv: The notation-key completeness table, one row per sector-category.
- pakemis_air_coverage_matrix.csv: The 700-cell coverage matrix (10 periods x 10 modules x 7 core species) under the closed IPCC notation keys.
- pakemis_air_source_coverage.csv: Process by species by period coverage.
- pakemis_artifact_inventory.csv: Every file this release ships, with the role it declares and who names it.
- pakemis_completeness_mass.csv: Unestimated streams joined to available activity-based screening quantities, with overlap and scenario assumptions retained.
- pakemis_coverage_matrix.csv: Records: coverage matrix.
- pakemis_coverage_summary.csv: Records: coverage summary.
- pakemis_national_series_coverage.csv: Records: national series coverage.
- pakemis_product_catalog.csv: One row per logical product, not per file.
- pakemis_product_coverage.csv: Records: product coverage.
- pakemis_product_facts.csv: One authoritative answer per fact a manuscript states about the shape of this release.

Activity accounts, source controls and extension

- ogra_gas_class_series.csv: OGRA's sector-wise gas classes per fiscal year (MMCFD, average daily).
- pakemis_agriculture_basis_change.csv: Source/basis change from archived FAOSTAT-level/calendar activity to current fiscal crops/nutrients and adopted-anchor livestock reconstruction.
- pakemis_agriculture_fiscal_series.csv: Agriculture fiscal activity, FY2015-16 to FY2024-25.
- pakemis_ammonia_nitric_series.csv: Ammonia (2.B.1) and nitric acid (2.B.2) as nine-year series.

- pakemis_brick_activity_reconstruction.csv: Brick-kiln coal from A kiln count and A measured fuel rate.
- pakemis_brick_coal_alternatives.csv: Brick-kiln coal - the bounded alternatives, per fiscal year.
- pakemis_cement_activity_sources.csv: Records: cement activity sources.
- pakemis_cement_kiln_coal_reconstruction.csv: Cement kiln coal reconstructed from clinker Q = K x H x s / NCV as a band per fiscal year.
- pakemis_coal_allocation_cases.csv: Dated conditional coal allocation cases.
- pakemis_coal_balance.csv: National coal balance per fiscal year supply lanes vs HDIP's allocation vs NEPRA's.
- pakemis_coal_change_decomposition.csv: Coal emissions, year on year.
- pakemis_coal_source_reconciliation.csv: Industrial coal - source reconciliation one row per fiscal year x HDIP consumer.
- pakemis_coal_vintage_ledger.csv: Consecutive prints of the same reference period, paired.
- pakemis_extension_activity_revisions.csv: Source availability, revisions and controls for common core and candidate extension.
- pakemis_extension_source_availability.csv: Source availability, revisions and controls for common core and candidate extension.
- pakemis_extension_source_controls.csv: Source availability, revisions and controls for common core and candidate extension.
- pakemis_fuel_sulfur_decision.csv: The fuel-sulfur decision record (item 3), under the same acceptance policy as.
- pakemis_fy2025_26_provisional_activity.csv: Separate provisional source activity and emissions.
- pakemis_fy2025_26_provisional_emissions.csv: FY2025-26 source estimates with explicit partial coverage.
- pakemis_hfc_bank_series.csv: HFC emissions from the household equipment bank, FY2015-16..FY2023-24.
- pakemis_industrial_change_attribution.csv: Where the 1.A.2 decline comes from.
- pakemis_industrial_coal_bridge.csv: Industrial coal - the replacement bridge and reference decision per fiscal year x component.
- pakemis_industrial_coal_decision.csv: The decision record under the acceptance policy.
- pakemis_industrial_screening_series.csv: The remaining industrial categories as nine-year screening series.
- pakemis_industry_boundary_ledger.csv: The industrial boundary ledger.
- pakemis_minerals_controls.csv: Independent source controls checked against final exported monthly/provincial panel.
- pakemis_ocac_extension_revisions.csv: Later publication overlap and unresolved sector by product revisions.
- pakemis_power_extension_controls.csv: Physical station versus system fuel and generation controls.
- pakemis_rice_geography_crosscheck.csv: The declared MapSPAM rice surface measured against the own-year GCD-Rice 2023 30 m classification, as within-province ADM2 shares with per-unit.
- pakemis_series_splice_agriculture.csv: Records: series splice agriculture.
- pakemis_sulfur_adoption_effect.csv: What the sulfur adoption moved, per stream and per year, with every sign shown.

Annual records

- national_all_sectors.csv: Ten sectors on one reporting period, exact mass closure and executable cross-boundary ledgers.
- national_all_sectors_ledger.csv: Records: national all sectors ledger.
- pakemis_air_annual_fy2015_16_fy2024_25.csv: Seven core pollutants in native mass with narrower additional species.
- pakemis_air_atomic_annual.csv: The ten-fiscal-year air panel's atoms.
- pakemis_air_power_omission.csv: What the covered national total leaves out at power stacks, per period and species.
- pakemis_air_source_classes_fy2015_16_fy2024_25.csv: The source-class air layer (industry, ippu, fugitives) at class grain, ten years.
- pakemis_annual_fy2015_16_fy2024_25.csv: Sector, category and gas for all 100 sector-years.
- pakemis_atomic_annual.csv: The component grain beneath the panel.
- pakemis_atomic_annual_aggregates.csv: Records: atomic annual aggregates.
- pakemis_claim_matrix.csv: Records: claim matrix.
- pakemis_deposit_manifest.csv: Every file the archive deposit ships, with the licence under which it ships.
- pakemis_dung_residual_ledger.csv: The unresolved dung residual (configuration S) by species.
- pakemis_facility_registry.csv: Records: facility registry.

- `pakemis_flatness_crosswalk.csv`: The crosswalk between the TE-class flat share and the physical flatness decomposition.
- `pakemis_ghg_configuration_atoms.csv`: Native GHG emissions for S_TAN, L_TAN, S_IPCC and L_IPCC.
- `pakemis_national_series.csv`: Records: national series.
- `pakemis_nitrogen_no_pathways.csv`: Agricultural NOx by pathway.
- `pakemis_paper_artefact_hashes.csv`: Records: paper artefact hashes.
- `pakemis_power_profile_variant.csv`: Records: power profile variant.
- `pakemis_power_station_status.csv`: Coal stations carrying each period's coal geography, joined to GEM unit start.
- `pakemis_source_register.csv`: Every registered source: whether the deposit ships it, and if not, how to obtain it.
- `pakemis_source_streams.csv`: Records: source streams.

Release provenance. This document describes PakEmis under the period, configuration and source boundaries specified in its methods. The identifiers below link it to the accompanying products and source register.

Release version	`1.0.0rc69`
Commit	`d8d874622f9a`
Reporting period	July 2015–June 2025 (ten fiscal years) (2015-07-01 to 2025-06-30)
Reference	S_TAN (household survey; TAN nitrogen)
FY2023–24 total	444.61 Mt CO_2e (AR5) under S_TAN, the reference; 449.85 Mt under L_TAN
Release manifest	`ac0ffe93ad9fcd88` (SHA-256 of `outputs/release_manifest.json`, first 16)
Source manifest	`c1be6d314cfce1ea` (SHA-256 of `data/manifests/source_manifest.csv`, first 16)
Data-description article	`paper_national/` (ESSD, in preparation), same release

References

Crippa, M., Solazzo, E., Huang, G., Guizzardi, D., Koffi, E., Muntean, M., Schieberle, C., Friedrich, R., and Janssens-Maenhout, G.: High resolution temporal profiles in the Emissions Database for Global Atmospheric Research, Scientific Data, 7, 121, https://doi.org/10.1038/s41597-020-0462-2, 2020.

EEA: EMEP/EEA air pollutant emission inventory guidebook 2023, European Environment Agency, Copenhagen, https://www.eea.europa.eu/publications/emep-eea-guidebook-2023, 2023.

FAO: FAOSTAT: emissions totals, crops and livestock, fertilisers by nutrient, and forestry production and trade, Food and Agriculture Organization of the United Nations, Rome [data sets], https://www.fao.org/faostat/, 2025 snapshot used in the source inventory; catalogue checked 30 September 2026, 2025.

Global Energy Monitor: Global Cement and Concrete Tracker, plant-level data, July 2026 (standard copy V1), Global Energy Monitor [data set], licensed under CC BY 4.0, 2026.

IPCC: 2006 IPCC Guidelines for National Greenhouse Gas Inventories, edited by: Eggleston, H. S., Buendia, L., Miwa, K., Ngara, T., and Tanabe, K., Institute for Global Environmental Strategies, Hayama, Japan, https://www.ipcc-nggip.iges.or.jp/public/2006gl/, 2006.

IPCC: Climate Change 2013: The Physical Science Basis, Cambridge University Press, Cambridge, https://doi.org/10.1017/CBO9781107415324, 2013.

IPCC: 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories, edited by: Calvo Buendia, E., Tanabe, K., Kranjc, A., Baasansuren, J., Fukuda, M., Ngarize, S., Osako, A., Pyrozhenko, Y., Shermanau, P., and Federici, S., IPCC, Switzerland, https://www.ipcc-nggip.iges.or.jp/public/2019rf/, 2019.

IPCC: Climate Change 2021: The Physical Science Basis, Cambridge University Press, Cambridge, https://doi.org/10.1017/9781009157896, 2021.

Kaza, S., Yao, L., Bhada-Tata, P., and Van Woerden, F.: What a Waste 2.0: A Global Snapshot of Solid Waste Management to 2050, World Bank, Washington, DC, https://doi.org/10.1596/978-1-4648-1329-0, 2018.

Ministry of Finance: Pakistan Economic Survey 2023–24, Government of Pakistan, Islamabad, chapter 2, Agriculture, 2024.

Ministry of Finance: Pakistan Economic Survey 2025–26, Government of Pakistan, Islamabad, agriculture chapter and statistical annexes, https://www.finance.gov.pk/survey/chapter_26/2_Agriculture.pdf, 2026.

NEPRA: State of Industry Report 2024, National Electric Power Regulatory Authority, Islamabad, https://nepra.org.pk/publications/State%20of%20Industry%20Reports/State%20of%20Industry%20Report%202024.pdf, 2024.

NFDC: Fertilizer Review 2023–24, Publication No. 2/2024, National Fertilizer Development Centre, Government of Pakistan, Islamabad, https://nfdc.gov.pk/publications/Annual%20Fertilizer%20Review%202023-24.pdf, 2024.

NFDC: Annual Fertilizer Review 2024–25, National Fertilizer Development Centre, Government of Pakistan, https://mnfsr.gov.pk/SiteImage/Misc/files/AFR2024-25.pdf, 2025.

Pak-EPA and EPMC: National Study on Privatization of Solid Waste Management in Eight Cities of Pakistan, Pakistan Environmental Protection Agency, Islamabad, 1996.

PBS: Household Integrated Economic Survey 2018–19, Pakistan Bureau of Statistics, Islamabad, survey report and microdata, 2019.

PBS: Seventh Population and Housing Census 2023, Pakistan Bureau of Statistics, Islamabad, census results, 2023.

PBS: Agricultural Census 2024, Pakistan Bureau of Statistics, Islamabad, report collection, https://www.pbs.gov.pk/agriculture-census/, accessed 30 September 2026, 2026a.

PBS: Household Integrated Economic Survey (HIES) 2024–25, Pakistan Bureau of Statistics, Islamabad, https://www.pbs.gov.pk/household-integrated-economic-survey-hies-2024-25/, accessed 30 September 2026, 2026b.

WHO and UNICEF: Joint Monitoring Programme for Water Supply, Sanitation and Hygiene: Pakistan household country file, 2023 snapshot, https://washdata.org/data/downloads, 2023.